Prepared for referee review

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision R73 addresses the 11 October 2026 advisory report on R67. The original NBO paper is extended by qualification-optimal continuation readouts, exact primal–dual certificates, a predictable original-law cost theorem and four predeclared construction services. Every prior active section and adverse evidence block is retained.

1. RESPONSE TO THE REFEREE

1.1. *Neural Bellman Operators — Revision R73*

We thank the referee for identifying where the original NBO construction needed an attained fitting criterion, an informative economic comparison and a complete numerical cost account. We revise the same paper. Its title, author, economic primitives, constructive continuation, original Bellman optimum and broader historical applications are retained. The response addresses the advisory report on R67 dated 11 October 2026, at review commit 4b27b56f4514f6c1d77b69fe707f7202f8b06b19. The development base is the subsequent R72 source commit 5611b32978a2df5f32bff796f7dd209105cf28b0, which retains the completed R70 revision and the frozen intervening scientific records. The report is owner-commissioned advice, not an official decision of Econometrica.

The revision adds two proved results and an executed construction. The qualification-optimal readout theorem turns a verified residual-oscillation bound into a finite linear program with an exact primal–dual certificate. Backward fitting uses the continuation actually returned at the following date. The predictable original-law cost theorem then supports repeated inspection and optional stopping, while distinguishing a fixed complete controller's cost from the conditional average of changing controllers. Four predeclared constructions in the original nonlinear investment economy fit and qualify actual ReLU and conventional continuations. The

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independent checker recomputes every new action value and every optimization certificate by a different exact integration route.

These additions complete a further link in the original theorem–implementation–policy-cost chain. They do not retitle the paper as a generic learning framework, replace its model, delete adverse observations, or infer a neural ranking from a representation identity. All preceding active sections are compiled byte-identically once in the current main article, Technical Supplement or Numerical and Development Appendix. The prior wrappers and complete editions remain available for comparison.

1.2. *M1 / B1. The original NBO object and the additional contribution*

The constructed continuation, an optional action proposal and a retrieved Bellman witness are distinct objects. The original witness-producing continuation and affine–ReLU realization remain in the main theorem chain. The common verifier continues to certify actions proposed by both neural and conventional methods; its correctness is not described as an intrinsically neural advantage.

R73 places the attained continuation fit directly inside the original-law economic account. A verified continuous-action Bellman enclosure at each cover point, a known feature modulus and a coefficient box determine the residual-oscillation objective. The new theorem gives its exact finite optimization representation and a dual lower bound. The achieved readout enters the next backward step and the frozen-continuation policy-cost identity. Thus the fitting criterion has a proved connection to the actual implemented policy's original-optimum bound and to the range used for cost inference. This is stronger than giving a neural name to a classical policy or reporting a small action-label error. It remains a statement about a declared fixed feature class, not a theorem that every neural encoding beats a conventional one.

1.3. *M2 / B3. Attained fitting quality rather than a hypothetical training success*

The intervening completed revision supplies certificate-aware fixed-feature action readouts, matched conventional controls and measured objective gaps. Those results

are retained. R73 additionally constructs continuation readouts by minimizing the bound that uniform residual qualification actually uses. For K coefficients and N label nodes, the linear program has $2K + 2$ variables and $2N + 2K$ inequalities apart from variable bounds. Its dual is written explicitly, and every solver-proposed coefficient and probability multiplier is rationalized and checked exactly. A solver convergence message alone has no certification authority.

The new study was committed before its four services ran. It crosses two midpoint-cover resolutions with additive shallow ReLU and degree-four Bernstein bases, each having nine coefficients and the same coefficient limits. At each date, the continuation at the next date is the one actually constructed by that service. A bounded least-squares readout uses the identical feature matrix and fixed future labels as a fitting-objective ablation. It is not a separately propagated least-squares controller. The complete catalogue, all solver messages and exact achieved gaps are deposited, including any failure to meet the prespecified optimization tolerance.

The theorem is global for its finite, fixed-future readout problem. It does not assert joint multi-date optimality, unrestricted hidden-layer convergence or minimization of true policy regret. Those distinctions do not weaken the original objective; they state exactly which attained computation the proof certifies.

1.4. *M3 / B5. Validation independence and adaptive deployment*

The retained study separates predictor-independent validation contexts from contexts visited during deployment, and the intervening deployment theorem conditions on the actual history. R73 adds the corresponding distinction for policy-cost qualification. Before an episode, its complete controller pair, continuation and support must be fixed given the preceding history. The continuation can change predictably between episodes without changing the expected original-law cost difference of a fixed controller pair. Within an episode it remains frozen for the telescoping identity.

The new cost theorem does not require independent, identically distributed deployment contexts. It bounds the average conditional expected cost difference under the stated conditional laws. That average equals one fixed policy-cost contrast

only when the complete controllers, initial law and reset contract stay fixed. A program with a cache carried between episodes can have a changing conditional target despite an unchanged code identifier. The manuscript and ledger therefore require a new error-allocated epoch for a changed fixed target. History-dependent controller memory must be included in the policy description and in the uniform greedy-gap qualification. Neither changing contexts nor reused validation data are silently treated as new independent evidence for the final controller.

1.5. *M4 / B2. Prospective reuse, warm starts and complete accounting*

The retained sixty-four-block catalogue and its cold-start comparisons remain unchanged. R72 further executes predeclared fresh-query, exact-key and geometrically transported-witness modes, including invalidation, capacity repair and reload checks. The new paper retains their actual root counts, descendants and receipts rather than replacing the original first-fill cost by an optimistic cache-hit model. The finite-cover theorem keeps its explicit retention and tighter-fill premises; the bounded executed cache is not claimed to satisfy them automatically.

R73 separates the cost of fitting a continuation from the cost of deciding that it is economically acceptable. The new construction charges its exact conditional action evaluations, original continuous-action covers, whole-cell feasible actor selection, optimization, rationalization, ablations and records. The cost ledger retains every attempted numerical episode. A numerical failure receives the previously qualified full support interval and its actual work; it is not discarded after an outcome-dependent failure. A failed support proof, by contrast, stops qualification. The adoption corollary explicitly amortizes setup over a declared deployment count, not over the number of inference observations.

1.6. *M5 / B4. Precision at an economic margin and valid stopping*

The completed R70 study retains twelve original-law centered intervals that meet its predeclared equivalence margin $1/256$. All twelve contain zero. We preserve the positive equivalence conclusion, the lack of identified strict superiority, the

earlier wide cost intervals and the complete qualification costs. No old fixed-sample interval is relabeled as an anytime result.

R72's frozen-continuation identity provides an alternative to costly optimal-value centering. Its range is determined by verified residual oscillation and actual greedy gaps, with the terminal correction retained when the fitted terminal value is inexact. R73 now executes the missing backward continuation fit and its uniform qualification. The exact primal–dual gap measures optimization of that stated qualification objective, while the published residual and regret bounds report its actual numerical conservatism.

The predictable cost theorem gives a new stopping-valid interval for verified continuous-law cost endpoints. A finite collection of predeclared bets and rational upper logarithm bounds makes it computable. It supports repeated inspection without spending the error budget again at every look. A finite bet set can retain an asymptotic radius; the supplement states that limitation and explains how a preallocated countable family can be used. The four new construction services add no independent policy-cost samples and do not claim to have beaten the old 1/256 qualification cost. Their positive result is an achieved, independently checked fit-to-range-to-policy-bound chain in the original economy.

1.7. *M6 / B6. Multiple controls and long-horizon computation*

The retained vector catalogue contains a genuine two-control feasible simplex, triangular lower certificates, conventional refinement and trained two-output proposals under the original continuous innovations. The four-date transferred model remains identified as a transferred model, not a newly trained four-date controller. The R72 query experiment further tests repeated witness use at two, three and four dates. Each miss retains the full original-law descendant cost.

The new qualification-readout and predictable cost theorems are not restricted to scalar actions: their premises are a valid original continuous-action enclosure, feature and Bellman moduli, and a qualified complete controller. The four new numerical constructions intentionally use the original two-state, scalar-control economy so their entire action catalogue can be checked independently from rational

primitives. They do not substitute for the existing vector evidence or assert a new high-dimensional scaling result. Their tensor state-cover cost is explicit. The non-tensor action verifier and the witness cache remain separate mechanisms with their original scope and cost accounts.

1.8. *M7 / B7. Independent numerical checks and the archive dependency closure*

R73's independent checker does not call the production conditional-moment expansion or the LP verifier. It evaluates terminal action values using the retained original rational primitive. For fitted ReLU continuations it partitions the innovation interval at all neuron breakpoints and integrates the affine pieces; for the degree-four Bernstein continuation it uses the positive five-point Boole formula, which is exact here. It reconstructs every continuous-action lower cover, robust feasible choice, whole-cell greedy-gap allowance, residual range, dual lower bound and accumulated policy bound. Replaying a hash alone is not this calculation.

The existing independent witness-transfer and polygon-integration checks remain, as do the prior whole-bin cost reconstruction and cumulative-buffer aliasing counterexample. The new proof does not independently recreate every descendant of the historical largest Bellman tree, and the release does not say otherwise. The checker boundary remains explicit for each evidence block.

We also located the R72 publication interruption. Its ordinary build had passed, but its clean archive omitted a transitive R62 numerical input imported through the recursive query code. The new archive contains the dependency closure, including the retained service49.py input. The repair preserves the old source and failure record; it does not remove the test or alter an inherited scientific result. The current documents and tables are rebuilt from an archive after generated outputs are removed, and the reproduction record compares their actual text, pages and hashes.

1.9. *M8 / B8. Reliability, clocks and statistical observations*

Every historical service, warning, adverse comparison and stopped pilot is retained. The new fixed-design construction has one complete service per declared design point. Its times are descriptive observations, not sampling estimates of mean

runtime or a cross-hardware speed theorem. Its operation counts, exact rational checks and fitting statuses are reported separately. Optimization accuracy is certified by its achieved gap, not by the absence of warnings. The independent checker records its own time without replacing the original service clock.

The exact test suite includes primal–dual fixtures, whole-domain residual checks, retained primitive comparisons, conditional-moment identities, target changes, failed numerical episodes and the full 4,096-leaf adaptive finite-model diagnostic. This diagnostic is a correctness regression, not a new investment simulation or an empirical proof of concentration. The probability statement is supplied by the theorem under its conditional-law assumptions. The release continues to distinguish code tests, scientific construction services, query workloads and independent policy-cost rows.

1.10. *M9 / B9. The economic decision and its units*

The economic object remains the actual feasible investment controller and the cost of constructing, qualifying and replacing it. The new stopping theorem permits a well-defined decision: stop qualification when the upper incremental-cost endpoint, plus an explicitly priced setup charge per declared deployment and a per-deployment charge, is negative. Equivalence at a specified loss band is a separate decision from strict improvement. New controller pairs can be admitted in conditionally budgeted epochs.

Measured processor seconds are not silently assigned a monetary value. The deployment count over which setup is amortized is not automatically the number of qualification episodes. A cached controller does not inherit the signed cost interval of a different fresh-search policy simply because both have the same regret allowance. These conditions make the original adoption question operational without inventing a calibrated installation fee, an estimated welfare effect or a specifically neural purchase recommendation. The existing economic-choice and information-acquisition results remain unchanged in the current reading set.

1.11. *M10 / B10. Exposition, literature and preservation*

The main article now connects the original construction and verifier to a fitted continuation, an attained qualification objective, stopping-valid policy costs and paid reuse. Its main new theorems state the full premises and conclusions; the Technical Supplement gives complete proofs. The previous optimal-advantage centering section moves intact into that supplement. The Numerical and Development Appendix retains the original training, localization, deployment, economic and empirical sections and adds the detailed new records. Every preceding active section is hash checked and assigned once; the old wrappers and broader complete editions are preserved.

The literature discussion distinguishes established safe policy iteration, convex approximation, value-function control variates, learned warm starts and time-uniform concentration from the present numerical contracts. The additional primary references are Howard and coauthors on nonnegative-supermartingale concentration and Karampatziakis, Mineiro and Ramdas on off-policy confidence sequences. We do not claim priority for those general principles. The contribution is the constructive readout dual and its link to verified original-law NBO observations, acquired feasibility, predictable stopping and a fully stated economic cost account.

The internal main-article and supplement page targets are publication discipline, not purported official Econometrica limits. We retain the journal class, author-year references, theorem statements, complete proofs and compact economic interpretation. This remains a substantial original-paper revision rather than a claim that moving an appendix alone satisfies every editorial length preference.

1.12. *Verification and renewed review*

The final release is conditioned on the source freeze, complete recorded scientific catalogue, independent exact audit, all inherited and new tests, section and evidence preservation, ordinary editable-source compilation, clean archive reproduction and visual inspection of the actual PDFs. The review-ready branch is created only after those checks. Failed earlier attempts remain recorded. Neither compilation nor an exact regression suite is described as an external referee's mathematical acceptance.

The submitted positive result is the original NBO construction extended by an achieved qualification-optimal continuation fit and a stopping-valid original-law cost decision, with full numerical and economic premises. All adverse rankings and unresolved comparative claims remain visible, so renewed review can assess the new mathematics and computation without a change of subject or an invented empirical victory.